CSC 372/472: Mobile Applications Development for Android

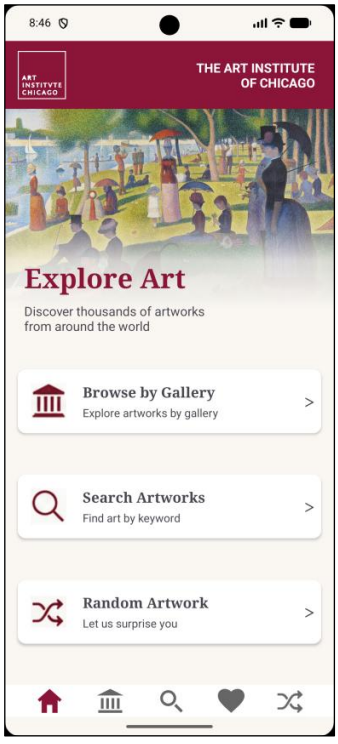
Assignment 3 – Art Institute API App *(300 pts)*

Uses: Internet, RecyclerView, Android Volley, Cloud API's, Glide, JSON Data, AlertDialogs, Browser Implied Intents, Custom Fonts

1. App Highlights:

- This application allows the user to view the galleries and artwork (both images and details) from the collection of the Art Institute of Chicago.
- Functionality includes browse artwork by gallery, view artwork details, search for artwork, mark artwork as favorites, and select a random artwork.
- Activities include MainActivity, GalleryActivity, ArtworkListActivity, ArtworkDetailActivity, SearchActivity, and FavoritesActivity
- You will need to use a different layout for landscape orientation for all application activities.
- The icons and images that are part of the application are provided to you. All other content comes from the Art Institute of Chicago's API.
- The application's activities are shown in detail in subsequent sections of this document.
- This application will use the Art Institute of Chicago's API. While there are a large number of endpoints and data values available, the specific API endpoints and data values needed to implement this application are detailed later in this document.

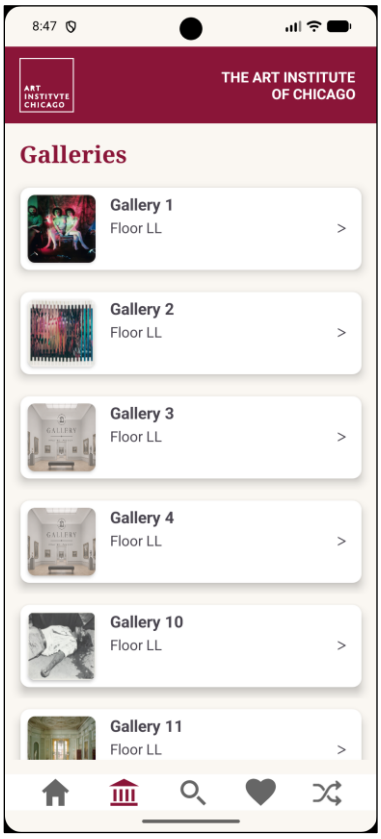
2. Application Activities:



MainActivity

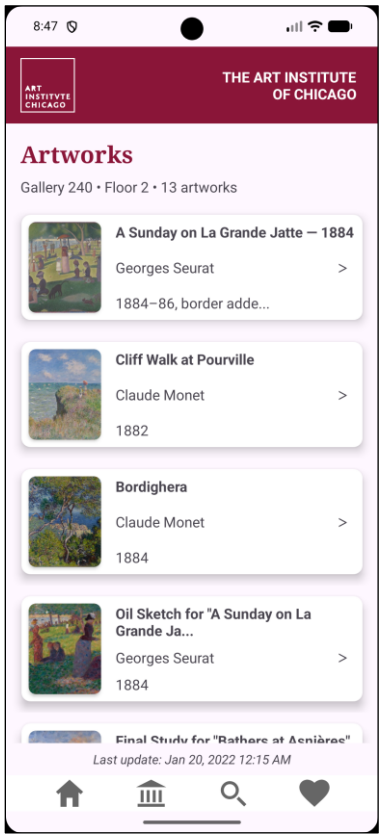
- This application allows the user to view the galleries and artwork (both images and details) from the collection of the Art Institute of Chicago.
- From the *MainActivity*, the user can Browse by Gallery, Search Artworks, or View a Random Artwork.
- The navigation icons along the bottom of the screen allow the user to return to the home screen, browse the galleries, search for Artwork, view favorites, and view a random artwork.
- Selecting the Home icon returns the user to the home screen from any other screen.
- Selecting the Gallery icon opens the galleries activity.
- Selecting the Search icon opens the search activity.
- Selecting the Favorites icon opens the favorites activity.
- Selecting the Random icon opens artwork activity to a randomly selected artwork.

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GalleryActivity

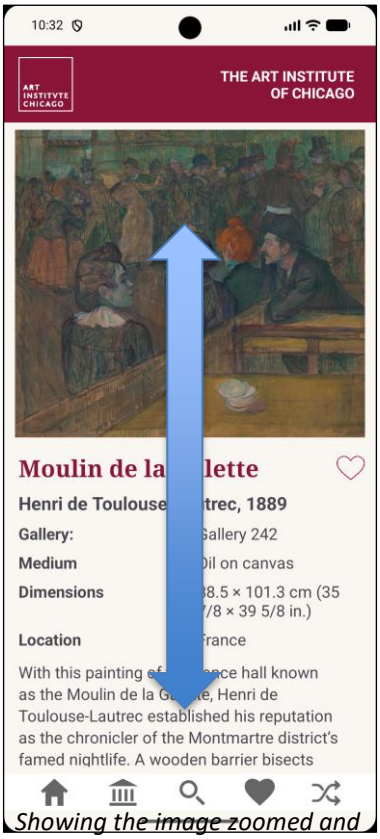
- This activity displays all galleries in a RecyclerView, showing the name, floor, and an image for each gallery. *Note: the image for each gallery is the first artwork returned for the selected gallery by the API.*
- Most gallery names are simply their gallery number: “Gallery 262”. Some galleries have actual names: “North Garden”. Simple use whatever name is returned by the API.
- Most galleries have a floor specified (“Floor 2”, “Floor 3”, etc). However some may not indicate a floor – the floor value should be left blank in the RecyclerView entry.
- Tapping on a gallery opens the Artworks activity, displaying all artworks in the selected gallery.
- The navigation icons along the bottom of the screen allow the user to return to the home screen, browse the galleries, search for Artwork, view favorites, and view a random artwork.
- When viewing the gallery activity, the gallery icon in the icons along the bottom of the screen will be red (rather than grey).



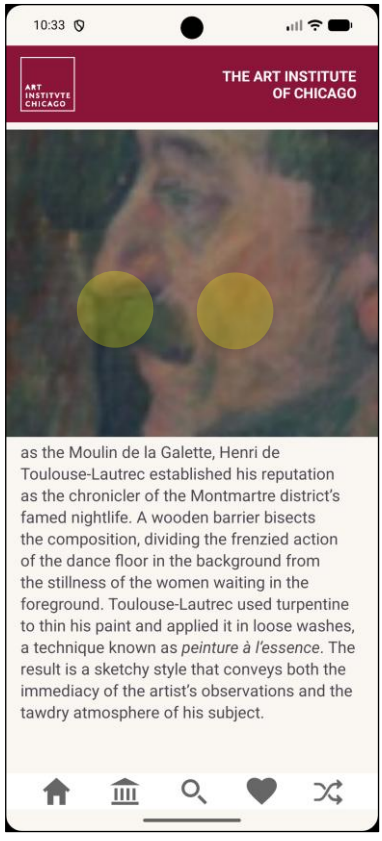
ArtworkListActivity

- Tapping on a gallery opens this artworks list activity, displaying all artworks in the selected gallery in a RecyclerView.
- The artworks list activity displays the gallery, the floor, and the number of artworks present (as seen in the provided image).
- Each entry shows the artwork name, the artist name, and the date for each artwork. *Note: The date can be a variety of values so just treat that field as a string, not a numeric value.*
- The “Last update” field seen below the RecyclerView is a value provided by the API.
- Tapping on an artwork opens the artwork detail activity, displaying the artwork and its related data content.
- The navigation icons along the bottom of the screen allow the user to return to the home screen, browse the galleries, search for Artwork, view favorites, and view a random artwork.
- When viewing the artwork list activity, all icons along the bottom of the screen will be grey.

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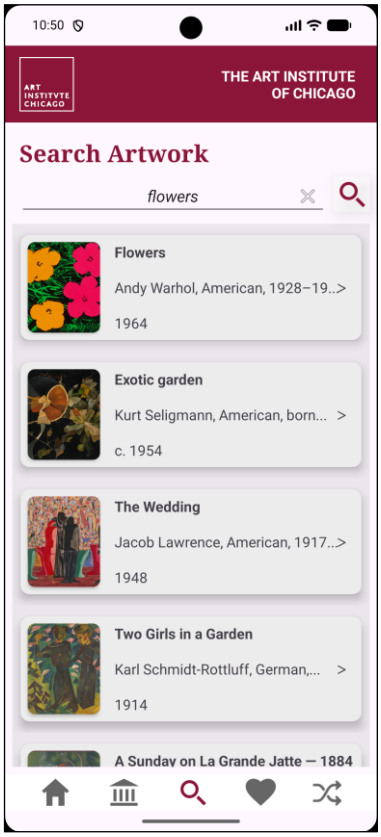
Showing the image zoomed and the information content scrolled.



ArtworkDetailActivity

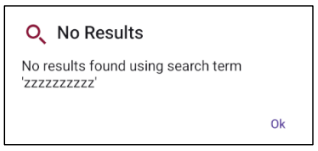
- Tapping on an artwork opens the artwork detail activity, displaying the artwork and its related data content.
- The content shown includes an image of the artwork at the top. This image should be a zoomable image (double tap or pinch-zoom to zoom).
- The data content, from the artwork image down through the artwork description (not including the navigation icons), is highly variable in length and should be put in a ScrollView so the data content can be scrolled if needed.
    - Below the image is the artwork name. The name can sometimes include extra content – like a date. Just use whatever content the API provides.
    - To the right of the name is the favorite icon (heart) The favorite icon is a heart outline ♡ if the artwork is not a favorite. It is a filled in heart ♥ if it is a favorite.
    - Below these is a combined value of the artist's name and the artwork date.
    - Below that comes the gallery name, the artwork medium, the dimensions, and the location (with labels as shown).
    - Lastly is the artwork description.
- Any missing data content (missing medium, missing location, etc.) should not be displayed (including the related labels).
- The navigation icons along the bottom of the screen allow the user to return to the home screen, browse the galleries, search for Artwork, view favorites, and view a random artwork.
- When viewing the artwork detail activity, all icons along the bottom of the screen will be grey.

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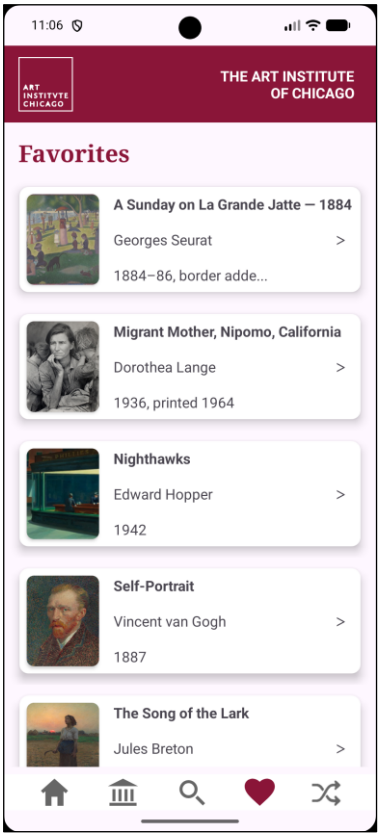


SearchActivity

- Selecting Search Artworks from MainActivity or tapping the search icon in the navigation icons along the bottom of the screen opens the search activity, allowing users to search for artworks by keyword(s).
- Keyword(s) can be entered as shown in the image. Tapping the search icon will invoke the API's search endpoint. The results are displayed in a RecyclerView as shown.
- If the search returns no results, an alert dialog should be displayed indicating no results were found:

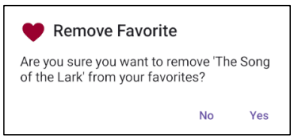


- The navigation icons along the bottom of the screen allow the user to return to the home screen, browse the galleries, search for Artwork, view favorites, and view a random artwork.
- When viewing the search activity, the search icon in the icons along the bottom of the screen will be red (rather than grey).



FavoritesActivity

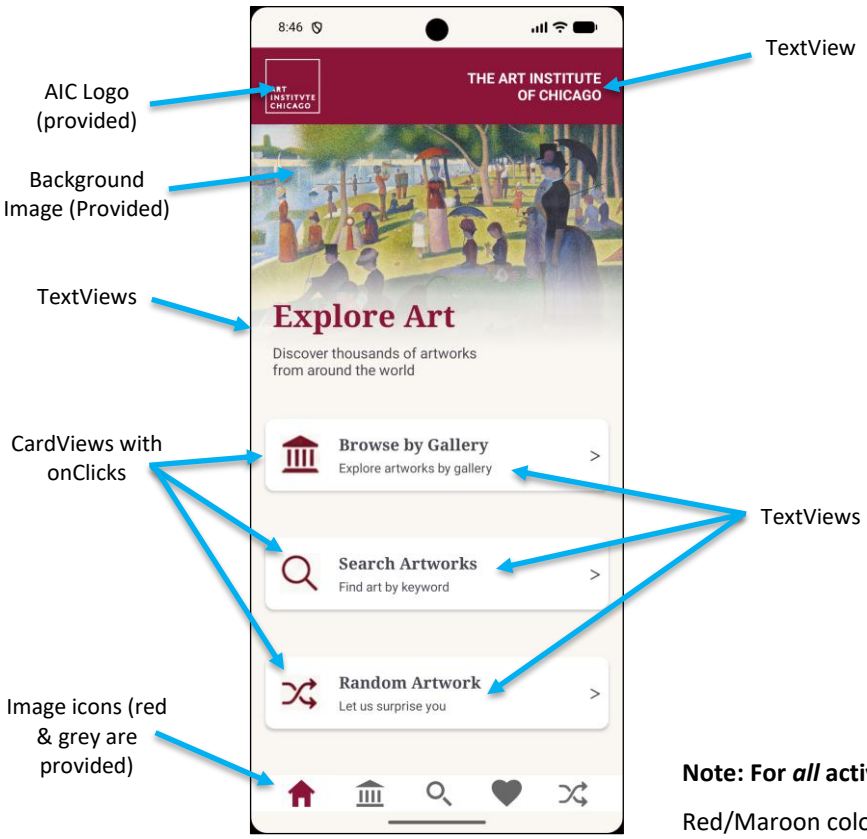
- Tapping the favorites icon in the navigation icons along the bottom of the screen opens the favorites activity, displaying all artworks the user has marked as a favorite.
- Each entry shows the artwork name, the artist name, and the date for each artwork. *Note: The date can be a variety of values so just treat that field as a string, not a numeric value.*
- Favorite artworks should be displayed sorted by the artwork name.
- Tapping on a favorite artwork opens the artwork detail activity, displaying the artwork and its related data content.
- Long-pressing on a list entry will remove the artwork as a favorite.
- Before removing, you should first display a confirmation dialog to confirm the removal:



- The navigation icons along the bottom of the screen allow the user to return to the home screen, browse the galleries, search for Artwork, view favorites, and view a random artwork.
- When viewing the favorites activity, the favorites icon in the icons along the bottom of the screen will be red (rather than grey).

3. Application Activity Details:

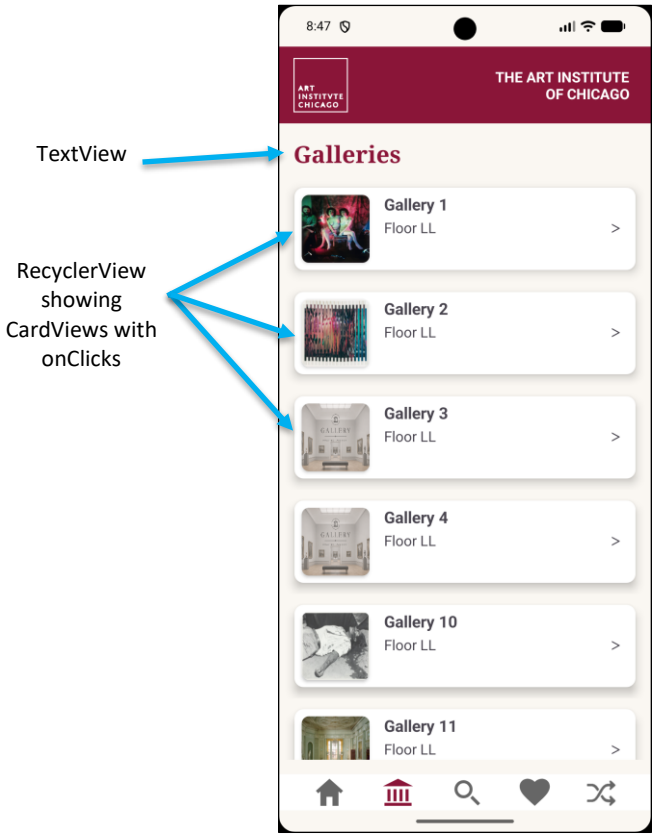
MainActivity



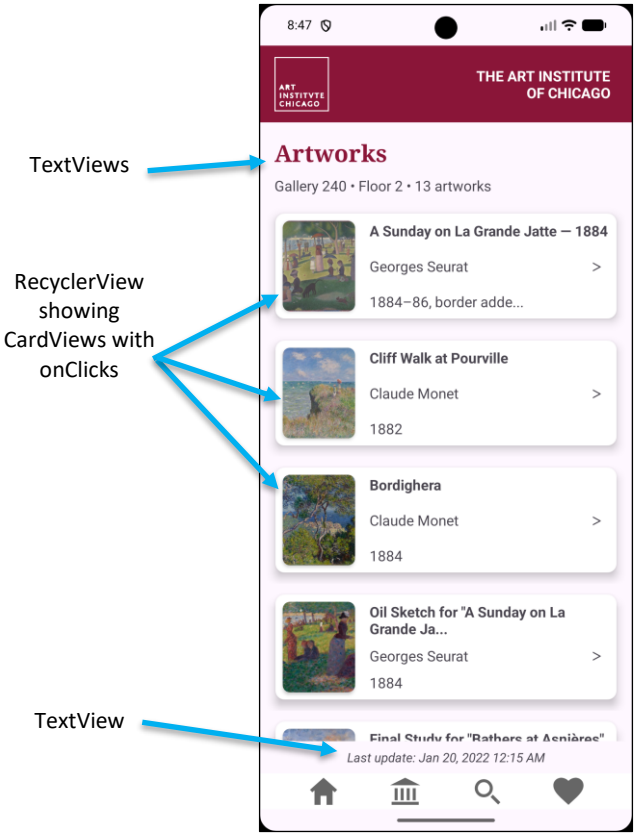
Note: For *all* activities:

Red/Maroon color:	#FF641027
Activity Background:	#FFFAF7F2
CardView Background:	#FFFFFFFF
"aic_grey" (used in icons):	#FF666666

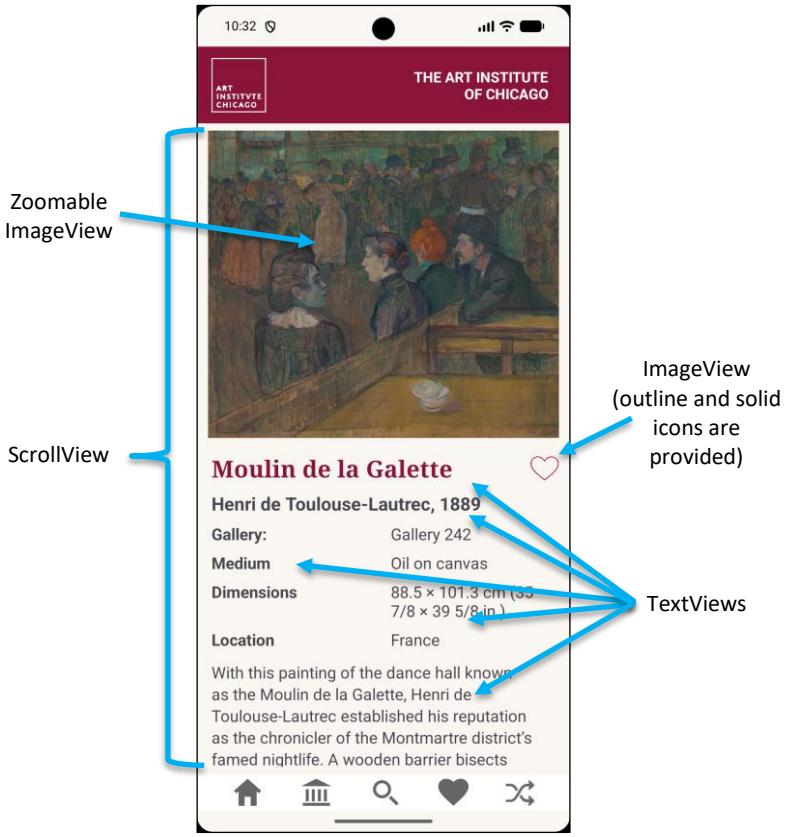
GalleryActivity



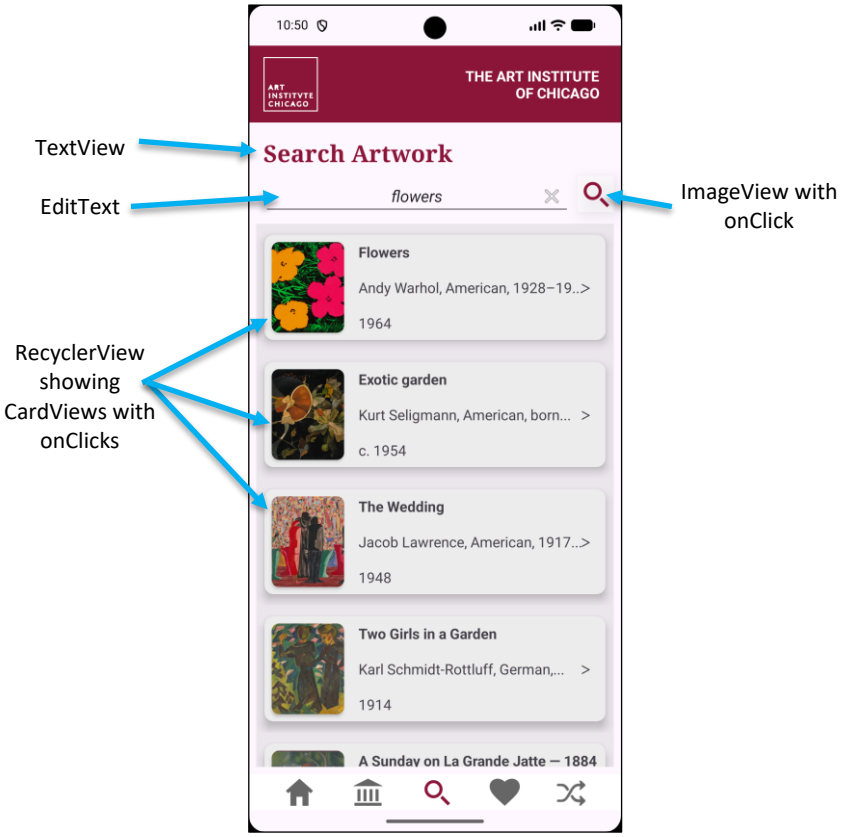
ArtworkListActivity



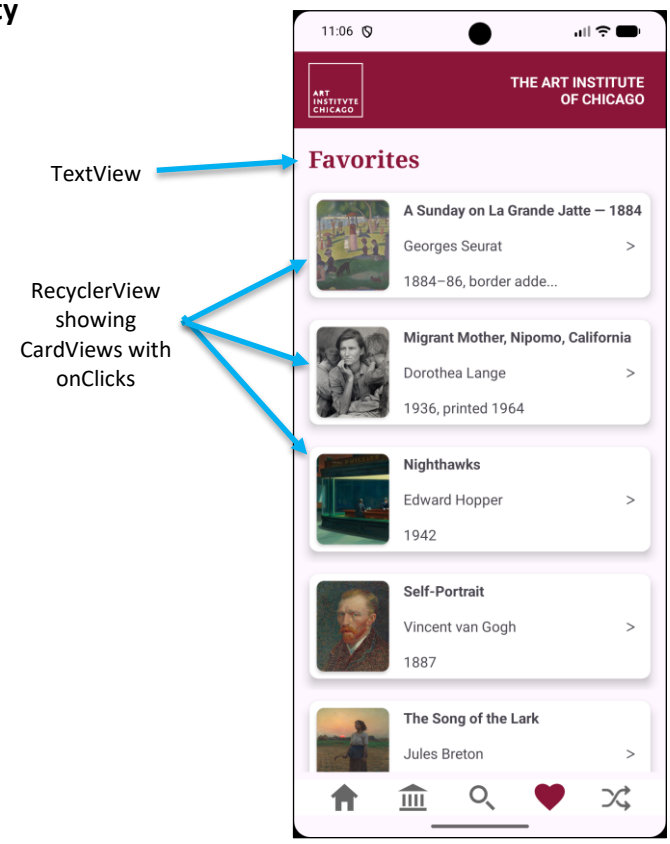
ArtworkDetailActivity



SearchActivity



FavoritesActivity



4. Landscape Layouts:

This application will need landscape layouts for each activity. You are to design and create these layouts on your own. Be sure all content is present/visible and usable in your landscape layouts.

These activities will need new layouts, as the original portrait layout is not usable in landscape orientation:

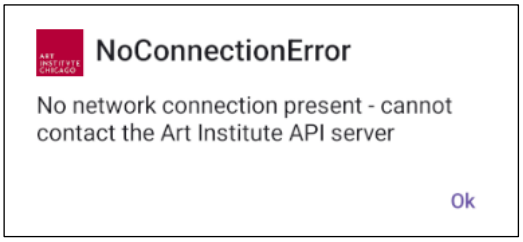
- MainActivity
- ArtworkDetailActivity

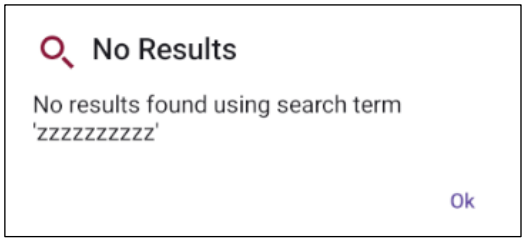
These activities will need new layouts, as while the original layout may work, you will only be able to see gallery or artwork at a time.

- GalleryActivity
- FavoritesActivity
- ArtworkListActivity
- SearchActivity

5. Error Conditions:

There are a variety of error conditions you should handle by displaying an AlertDialog to the user indicating what went wrong. For example:





You should trap and handle any error conditions you discover in this fashion. No error situation should generate log messages only if the error in some ways affects the display of application content.

6. Cloud-Based API Data:

Data will be obtained from the Art Institute via the Art Institute of Chicago API (https://api.artic.edu/docs). While there are many useful endpoints offered by this API, we will be using relatively few, as described below. The entire app can be done using these 2 endpoints.

1) Gallery Query (get all galleries)

Base endpoint URL: `https://api.artic.edu/api/v1/galleries`

Query parameters:

`fields` Data fields (comma separated) to return for each artwork in the results.
Use this entire string: id,title,number,floor,is_closed,source_updated_at

`limit` Number of search results to return per response "page".
Use: 100. This should be a string resource, not hard-coded

`page`: Specifies the "page" number of results to retrieve.
Use: Start with 1 *(you will need to pull multiple pages to get the full gallery list)*

Example complete URL:

```
https://api.artic.edu/api/v1/galleries?fields=id%2Ctitle%2Cnumber%2Cfloor%2Cis_closed%2Csource_updated_at&limit=100&page=1
```

Example results: *(Data of interest is in **bold**)*

```
{
  "pagination" : {
    "total" : 179,
    "limit" : 100,
    "offset" : 0,
    "total_pages" : 2,
    "current_page" : 1,
    "next_url" :
"https://api.artic.edu/api/v1/galleries?page=2&fields=id%2Ctitle%2Cnumber%2Cfloor%2Cis_closed%2Csource_updated_at&limit=100"
  },
  "data" : [ {
    "id" : 2147478068,
    "title" : "Gallery 272",
    "is_closed" : false,
    "number" : "272",
    "floor" : "2",
    "source_updated_at" : "2026-07-02T11:07:30-05:00"
  }, {
    "id" : 25290,
    "title" : "Gallery 234",
    "is_closed" : false,
    "number" : "234",
    "floor" : "2",
    "source_updated_at" : "2026-07-01T13:02:54-05:00"
  }, {
```

```
    ...
  }, {
    "id" : 23992,
    "title" : "Gallery 394",
    "is_closed" : false,
    "number" : "394",
    "floor" : "3",
    "source_updated_at" : "2022-04-15T16:40:14-05:00"
  } ],
  "info" : {
    "license_text" : "The data in this response is licensed under a Creative
Commons Zero (CC0) 1.0 designation and the Terms and Conditions of artic.edu.",
    "license_links" : [ "https://creativecommons.org/publicdomain/zero/1.0/",
"https://www.artic.edu/terms" ],
    "version" : "1.14"
  },
  "config" : {
    "iiif_url" : "https://www.artic.edu/iiif/2",
    "website_url" : "http://www.artic.edu"
  }
}
```


2) Artworks List Query (get all artwork in a gallery)

Base endpoint URL: `https://api.artic.edu/api/v1/artworks/search`

Query parameters:

`query[term][gallery_id]` Provide the gallery id (the id, not the name) here (i.e., `2147479701`)

`fields`: Data fields (comma separated) to return for each artwork in the results.

Use this entire string:
`id,title,artist_display,date_display,medium_display,place_of_origin,image_id,dimensions,short_description,description,credit_line,galleryId`

`limit`: Number of search results to return per response "page".

Use: 100. This should be a string resource, not hard-coded

`page`: Specifies the "page" number of results to retrieve.

Use: Start with 1 *(you will need to pull multiple pages to get the full gallery list)*

Example complete URL:

`https://api.artic.edu/api/v1/artworks/search?query%5Bterm%5D%5Bgallery_id%5D=2147479701&fields=id%2Ctitle%2Cartist_display%2Cdate_display%2Cmedium_display%2Cplace_of_origin%2Cimage_id%2Cdimensions%2Cshort_description%2Cdescription%2Ccredit_line%2CgalleryId&limit=100&page=1`

Example results: *(Data of interest is in **bold**)*


```
{
  "preference" : null,
  "pagination" : {
    "total" : 19,
    "limit" : 100,
    "offset" : 0,
    "total_pages" : 1,
    "current_page" : 1
  },
  "data" : [ {
    "_score" : 16223.775,
    "id" : 154235,
    "title" : "The Girl by the Window",
    "date_display" : "1893",
    "artist_display" : "Edvard Munch (Norwegian, 1863–1944)",
    "place_of_origin" : "Norway",
    "description" : "<p>Standing before a window in the dead of night, a young girl
                     draws back a curtain to gaze at the street from her darkened
                     room. Deep shadows envelop the interior, obscuring all but the
                     suggestion of a rounded piece of furniture at the lower right.
                     The darkness, combined with the sharply tilted floor, gives a
                     sense of precariousness and isolation to the figure. Loosely
                     applied brown, blue, and violet tones enhance the melancholy
                     atmosphere. The window serves as a symbolic barrier, separating
                     the interior from the outside world, but also as a membrane,
                     through which moonlight casts glowing rectangles upon the
                     floor. Although the girl’s actions suggest a narrative, the
                     painting only provides mystery—we cannot see her facial
                     expression, nor what she observes.</p>\n<p><em>Girl by the
                     Window</em> belongs to a group of nocturnal scenes painted by
                     Edvard Munch in the early 1890s, all characterized by moody
                     blue tonalities. Around that time he penned a manifesto in
                     which he declared that artists should depict “living people who
                     breathe and feel, suffer and love.” This focus on the
                     psychological realities of human existence found its ultimate
                     expression in Munch’s iconic work <em>The Scream</em> (1893;
                     Nasjonalmuseet, Oslo).</p>\n",
    "short_description" : "Edvard Munch’s life and art have come to epitomize
                     modern notions of anxiety. Created the same year as his
                     renowned <em>The Scream</em>, this painting depicts a young
                     girl in her nightgown gazing out at the city from her darkened
                     room. The steep angle of the floor and the deep shadows that
                     obliterate almost everything in the room create an unsettling
                     and enigmatic scene.",
    "dimensions" : "96.5 × 65.4 cm (38 × 25 3/4 in.); Framed: 110.5 × 79.4 × 8.9 cm
(43 1/2 × 31 1/4 × 3 1/2 in.)",
    "medium_display" : "Oil on canvas",
    "credit_line" : "Searle Family Trust and Goldabelle McComb Finn endowments;
Charles H. and Mary F. S. Worcester Collection",
    "image_id" : "d7df2633-3b40-f570-c906-211503a37cde"
```

```
  }, {
    ...
  }, {
    "_score" : 3.0000107,
    "id" : 260122,
    "title" : "Vase",
    "date_display" : "c. 1907",
    "artist_display" : "Taxile Doat (French, 1851–1938)\nSevres, France",
    "place_of_origin" : "Sèvres",
    "dimensions" : "34.8 × 21.5 cm (13 5/8 × 8 3/8 in.)",
    "medium_display" : "Glazed porcelain",
    "credit_line" : "Partial gift of Jason Jacques; Richard T. Crane Jr. Memorial
Fund",
    "image_id" : "69762997-20b5-6a73-bf7e-dde26c383b1a",
    "short_description" : null,
    "description" : null
  } ],
  "info" : {
    "license_text" : "The `description` field in this response is licensed under a
Creative Commons Attribution 4.0 Generic License (CC-By) and the Terms and
Conditions of artic.edu. All other data in this response is licensed under a
Creative Commons Zero (CC0) 1.0 designation and the Terms and Conditions of
artic.edu.",
    "license_links" : [ "https://creativecommons.org/publicdomain/zero/1.0/",
"https://www.artic.edu/terms" ],
    "version" : "1.14"
  },
  "config" : {
    "iiif_url" : "https://www.artic.edu/iiif/2",
    "website_url" : "http://www.artic.edu"
  }
}
```


7. Accessing artwork images

Artwork images can be accessed as follows:

For thumbnails (used in RecyclerViews):

```
String imageUrl =
      "https://www.artic.edu/iiif/2/" + imageId + "/full/400,/0/default.jpg";
```

For full-sized images (used in ArtworkDetailActivity):

```
String imageUrl =
      "https://www.artic.edu/iiif/2/" + imageId + "/full/1200,/0/default.jpg";
```

If the above request results in an error, try again using 800 size rather than the 1200 size.

If that also fails, display the default artwork image (provided) instead.

Use the Glide library to access the images at the indicated URLs. These image endpoints require User Agent headers as follows:

```
GlideUrl glideUrl =
        new GlideUrl( imageUrl,
                new LazyHeaders.Builder()
                        .addHeader("User-Agent", "ArticApp/1.0")
                        .addHeader("AIC-User-Agent", "ArticApp/1.0")
                        .build());
```

8. Supplied Resources

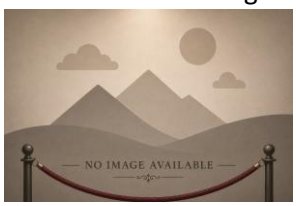
- **Images:**

AIC Logo (for toolbar)

Alert Dialog Logo



Default Artwork Image



Favorite Artwork (off)



Favorite Artwork (on) ← Used in artwork detail activity



Bottom navigation icons (these are xml drawable files)


ic_favorites.xml


ic_favorites_grey.xml


ic_galleries.xml


ic_galleries_grey.xml


ic_home.xml


ic_home_grey.xml


ic_search.xml


ic_search_grey.xml

Main activity image



Random icon



Random Icon (grey)



9. Fonts

No custom fonts are needed for this app.

- For bold and red labels, use the built-in font family "serif".
- For all other test, use the default font.

Assignment Assistance

If you are stuck on an assignment problem that you have exhaustively researched and/or debugged yourself, you can email me a ZIP file of your entire project so that I can examine the problem. All emailed assistance requests must include a detailed description of the problem, and the details of what steps you have already taken in trying to determine the source of the problem.

Note: To make your submission zip file smaller, before zipping your project file, you can remove the ".gradle" folder (found in your project's root directory), and remove the "build" folder (found in the "app" folder in your project's root directory).

Grading

- Base App *[75% of credit]*
  - Home Screen (MainActivity) *[10% of credit]*
  - Gallery Browsing (GalleryActivity) *[20% of credit]*
  - Gallery Artwork Browsing (ArtworkListActivity) *[20% of credit]*
  - Gallery Artwork Detail (ArtworkDetailActivity) *[25% of credit]*
- Additional Functionality (your choice – any of these, one, two or all three, in any order)
  - Search Activity (SearchActivity) *[10% additional credit]*
  - Favorites (Favorites Activity) *[15% additional credit]*
  - Random Artwork Detail (uses ArtworkDetailActivity) *[10% additional credit]*
- Credit:
  - Base App (75% * 300 pts = 225 pts)
  - Additional Credit:
    - Search (10% * 300 pts = 30 pts)
    - Favorites (15% * 300 pts = 45 pts)
    - Random (10% * 300 pts = 30 pts)
  - Total Possible
    - 75% + 10% + 15% + 10% = 110%
    - 225 pts + 30 pts + 45 pts + 30 pts = 330 pts

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Submissions

- Submissions must consist of your zipped project folder (please execute Build =>Clean Project before generating the zip file).
- Submissions should reflect the concepts and practices we cover in class, and the requirements specified in this document.
- Late submissions will be penalized by 10% per week late. (i.e., from one second late to 1 week late: 10% penalty). No submissions allowed after 1 week late.
- Grading will be based upon the presence and proper functionality of *all features and behaviors* described in this document.
- Grading will be performed with the following SDK details:
  - Compile & Target SDK Version: 37
  - Project Minimum SDK Version: 33
- Grading will be performed on emulator devices:

  Pixel 10 System Image API 37.0

  Pixel 7a System Image API 37.0

NOTE

This assignment is worth 300 points. ***(your previous assignment was worth 200 points)***

Note that this also means that the 10% late submission penalty will be 10% * 200 points = 20 points.

No submissions will be accepted beyond 4 days late

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